

EXHIBIT 36

Leakage Mechanisms in the Trench Transistor DRAM Cell

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Abstract—This paper discusses a band-to-band tunneling mechanism in the trench transistor cell (TTC), which is used in Texas Instruments' 4-Mbit DRAM. This effect should be operative in the class of trench cells where the charge is stored inside the trench and the substrate forms a capacitor plate. This effect does not compromise the functionality of the cell; in fact, it has the potential of improving the long-term reliability of the cell by preventing electrical overstress of the trench capacitor oxide.

I. INTRODUCTION

A 4-MEGABIT DRAM has been fabricated using a novel $9\text{-}\mu\text{m}^2$ one-transistor cell where both the pass transistor and the storage capacitor are incorporated in a trench [1]. This paper will discuss a band-to-band tunneling mechanism in the trench transistor cell (TTC). This effect should be operative in the class of trench cells where the charge is stored inside the trench and the substrate forms a capacitor plate.

II. CELL STRUCTURE

An $8\text{-}\mu\text{m}$ -deep trench ($1.3\text{ }\mu\text{m} \times 1.5\text{ }\mu\text{m}$) is etched using reactive ion etching in a $4.5\text{-}\mu\text{m}$ epitaxial layer on a p^+ substrate. A vertical pass transistor is made in the top $2\text{ }\mu\text{m}$ of the trench. The storage capacitor is also in the trench, below the pass transistor ($t_{\text{ox}} = 15\text{ nm}$). The inner plate of the capacitor is an n^+ polysilicon plug, and the outer plate is the p^+ substrate. The source of the transistor is connected to the n^+ polysilicon plug of the capacitor by a lateral contact that is made by an oxide undercut etch and refilling with polysilicon. The drain, gate, and source of the trench pass transistor are made of diffused-buried- n^+ bitline, n^+ polysilicon wordline, and a lateral contact, respectively [2], [3].

The TTC stores charge on the n^+ polysilicon plug inside the trench. Since the lateral contact is connected to the n^+ poly plug, one can envision the TTC capacitor structure as an MOS transistor with the gate, drain, and source tied together, which can never go into inversion at any gate bias. Hence, the storage capacitance is the series combination of the gate oxide capacitance and the deple-

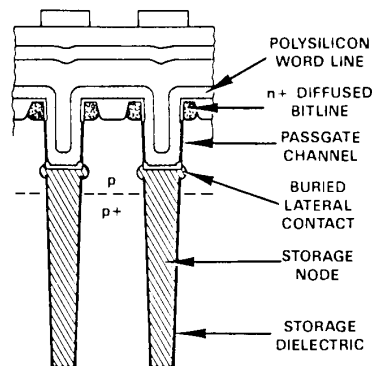


Fig. 1. TTC schematic diagram.

tion capacitances in the plates of the capacitor. Although the n^+ poly plug is heavily doped ($2E19\text{ cm}^{-3}$), the doping of the outer plate, which depends on updiffusion of boron from the p^+ substrate into the p^- epi layer, varies from 10^{19} cm^{-3} in the bottom of the trench capacitor to 10^{16} cm^{-3} near the lateral contact (Fig. 1).

III. CELL CHARACTERIZATION

The test structure used to characterize the TTC is shown in Fig. 2. It consists of a TTC and a conventional planar cell on the same bit line with a source follower to sense the output. The planar cell uses a $5\text{-}\mu\text{m}/1.5\text{-}\mu\text{m}$ planar pass transistor connected to an inversion-type planar storage capacitor (140 fF). The capacitor is $100\text{ }\mu\text{m}^2$ with a 25-nm gate oxide. The field plate of the storage capacitor is held at 5 V to invert the substrate underneath. The "0" level is set at 0 V while the "1" level is varied between 3 and 7 V . The bit line is precharged to $1/2 V_{cc}$ (2.5 V) and the word line switched between 0 and 8 V to transfer charge between the cell and the bit line while the substrate is held at -2.5 V . Typical output waveforms for a TTC, shown in Fig. 2, confirm the functionality of the cell.

Typical burst refresh data at temperatures between -50 and 100°C are shown for the TTC and the planar cell in Fig. 3(a) and (b), respectively. From a comparison of the output signals from the TTC and the reference planar cell, the trench cell capacitance is estimated to be about 22 fF . Refresh times at room temperature and 100°C are 1 s and 20 ms for the TTC. The corresponding values for the planar cell are 10 s and 100 ms . This suggests that there may be additional leakage mechanisms in the TTC compared to the conventional planar cell.

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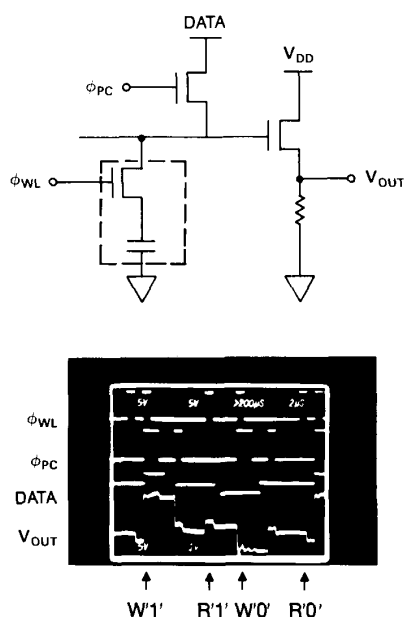


Fig. 2. Test circuitry to verify cell functionality and typical output waveforms from TTC.

To examine the leakage mechanisms of the trench cell further, the refresh characteristics of the TTC have been studied as a function of write “1” voltage at various temperatures (Fig. 4(a)). It is seen that the output signal does not increase linearly with the write “1” voltage as would have been expected, but tends to saturate for voltages above 5 V. This effect is more pronounced at higher temperatures and longer refresh times, indicating that there is a leakage current in the TTC that increases with temperature and is a strong function of the voltage in the cell. For comparison, the same test was performed on the planar cell. In this case, the output increases linearly with the “1” level (Fig. 4(b)). For the planar cell, the leakage currents are much lower than for the TTC and do not have as strong a dependence on voltage; hence, there is no “clamping” effect on the output voltage.

IV. POSSIBLE LEAKAGE MECHANISMS

Since the charge in the TTC is stored in the n^+ poly plug, which is completely oxide isolated except at the lateral contact, the charge in the cell can leak either through the capacitor oxide or through the lateral contact. At 5 V in the cell with -2.5 -V substrate bias, the field across the 150 -Å capacitor oxide is 5.7 MV/cm, which is barely enough for the onset of Fowler-Nordheim (F-N) tunneling through the oxide. However, from Gauss’ law, invoking continuity of the electric displacement across the Si-SiO₂ interface, the field in the p^+ Si near the interface is

$$E_{si} = \left(\frac{\epsilon_{ox}}{\epsilon_{si}} \right) E_{ox} = 1.9 \text{ MV/cm} \quad (1)$$

which is high enough for breakdown in the p^+ Si substrate. (The breakdown field in Si is 0.3 MV/cm.) How-

ever, for band-to-band tunneling in Si, not only must the field be high, but the surface potential band bending must also be in excess of the bandgap (1.1 eV), i.e., there must be filled valence band states and empty conduction band states very close (about 100 Å) to each other.

In an MOS capacitor, as the gate bias is increased such that the Si substrate depletes, the band bending can never exceed twice the Fermi potential, $2\Phi_F$, because of surface inversion. In the TTC, however, as mentioned before, the lateral contact (source) is tied to the n^+ poly plug (gate) and the capacitor can never go into inversion; instead, the substrate can go into deep depletion and the surface potential can bend in excess of the bandgap. Hence, electron-hole pairs can be generated by band-to-band tunneling in the p^+ Si; the electrons would then drift in the surface channel to the lateral contact and gradually discharge the storage node.

Another possible leakage mechanism is p-n junction leakage at the reverse-biased lateral contact. This could be either by diffusion of thermally generated minority carriers from the bulk or by generation-recombination in the depletion region because of traps associated with the poly plug and/or dislocations generated by stress in the Si in the vicinity of the trench. At higher voltages, there could be impact ionization and avalanching at the lateral contact, leading to the observed strong voltage dependence of the leakage. There could also be tunneling across the lateral contact-substrate p-n junction itself because of the high doping on both sides.

Yet another possibility is subthreshold leakage through the trench transistor from the cell to the bit line. However, data from the pass transistor [4] indicate that this leakage is less than 10 fA/cell and therefore is unimportant. A final possibility is cell-to-cell leakage through the buried lateral contacts.

V. EXPERIMENTAL LEAKAGE CHARACTERISTICS

The first possibility is F-N tunneling or some other leakage through the capacitor oxide itself. To test this, a trench capacitor structure was fabricated without the lateral contact where after the trench etch and capacitor oxide growth, the trench was refilled with the n^+ poly plug. The plug therefore was completely isolated by the storage capacitor oxide. The leakage was measured between the n^+ poly plug and the substrate. Leakage data for these structures is shown in (Fig. 5) for two different thermally grown oxide thicknesses. It is seen that the leakage current follows a (J/E^2) versus $(1/E)$ relationship, characteristic of F-N leakage. The F-N current density is given by [5]

$$J = \frac{q^3 E^2}{8\pi\hbar\phi} \exp \left[-\frac{4\sqrt{2m^*}\phi^{1.5}}{3\hbar q E} \right] \\ = AE^2 \exp \left[-\frac{B}{E} \right] \quad (2)$$

where $m^* = 0.42m_0$ is the effective tunneling electron mass in SiO₂, and ϕ is the barrier height for tunneling

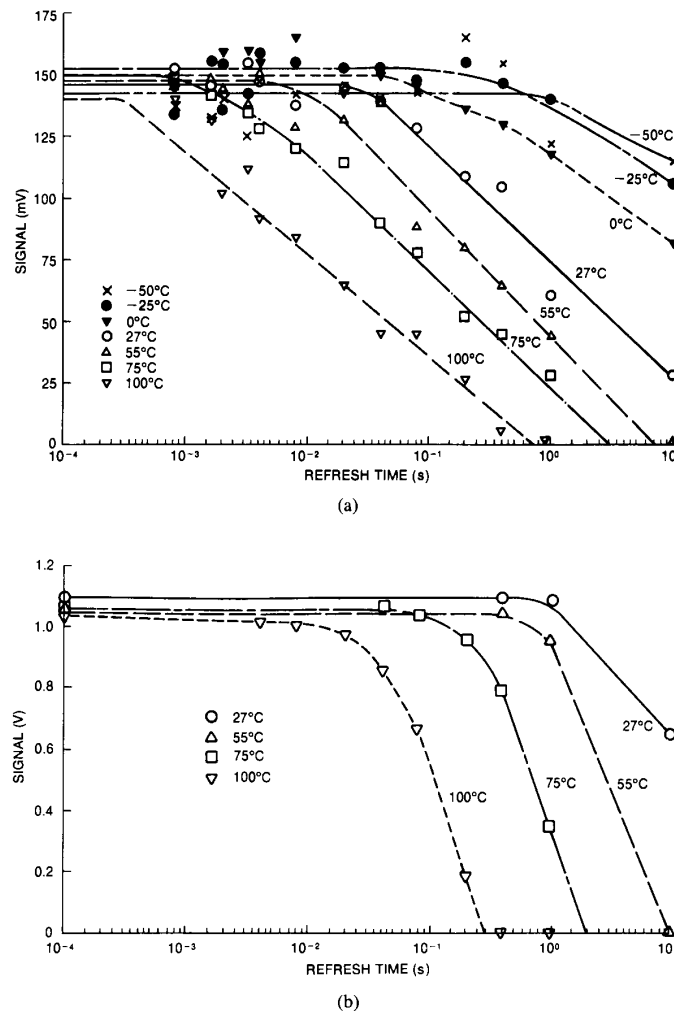


Fig. 3. Temperature dependence of refresh characteristics of (a) TTC; (b) planar cell. The "1" level is at 5 V and the substrate at -2.5 V.

equal to 3.2 eV for Si-SiO₂, which is the conduction band discontinuity at the Si-SiO₂ interface [5].

From Fig. 5, A and B are estimated to be $1\text{E-}7$ A/V² and $2.0\text{E}8$ V/cm, respectively. The coefficient " B " has been reported to be $2.5\text{E}8$ V/cm for Si-SiO₂ [6] and is important in the estimation of the barrier height. The discrepancy is probably due to the uncertainty in the trench capacitor oxide thickness. Although the oxides are nominally 150 or 200 Å (as measured on dummy slices), there is a range of oxide thickness in the trench partly because of different oxidation rates on the different crystal planes on the trench sidewalls, and partly because of the retardation of oxidation rate on concave surfaces (such as the bottom) of the trench [4], [7]. If the 150-Å (nominal) oxide is assumed to yield 120 Å in the thinnest regions, a value of $B = 2.5\text{E}8$ V/cm is obtained in closer agreement with published values. Uncertainties in the oxide thickness may also be responsible for the different slopes for the 150- and 200-Å cases.

From the measured value of B , a barrier height, $\phi = 3.1$ eV is estimated, which agrees with the theoretical value for the conduction band discontinuity at the Si-SiO₂ interface. Since the leakage measurements were taken with positive bias applied to the n⁺ poly plug, electrons must tunnel from the conduction band of the p⁺ substrate to the conduction band of the oxide. (The probability for hole tunneling is much less because of the greater barrier height.) The p⁻ substrate inverts in the low doping region, the inversion electrons being supplied by the buried-n⁺ bit line, and it is these inversion electrons that undergo F-N tunneling. However, the absolute magnitude of the leakage in the trench capacitor oxide is at least a factor of 100 lower than that observed in the TTC. Hence, this mechanism may be discounted in the actual TTC.

Experimental leakage characteristics for the TTC were obtained from a special test structure with 10 000 trench cells connected in parallel. The cells are accessed by turning the common word line high and applying bias to the

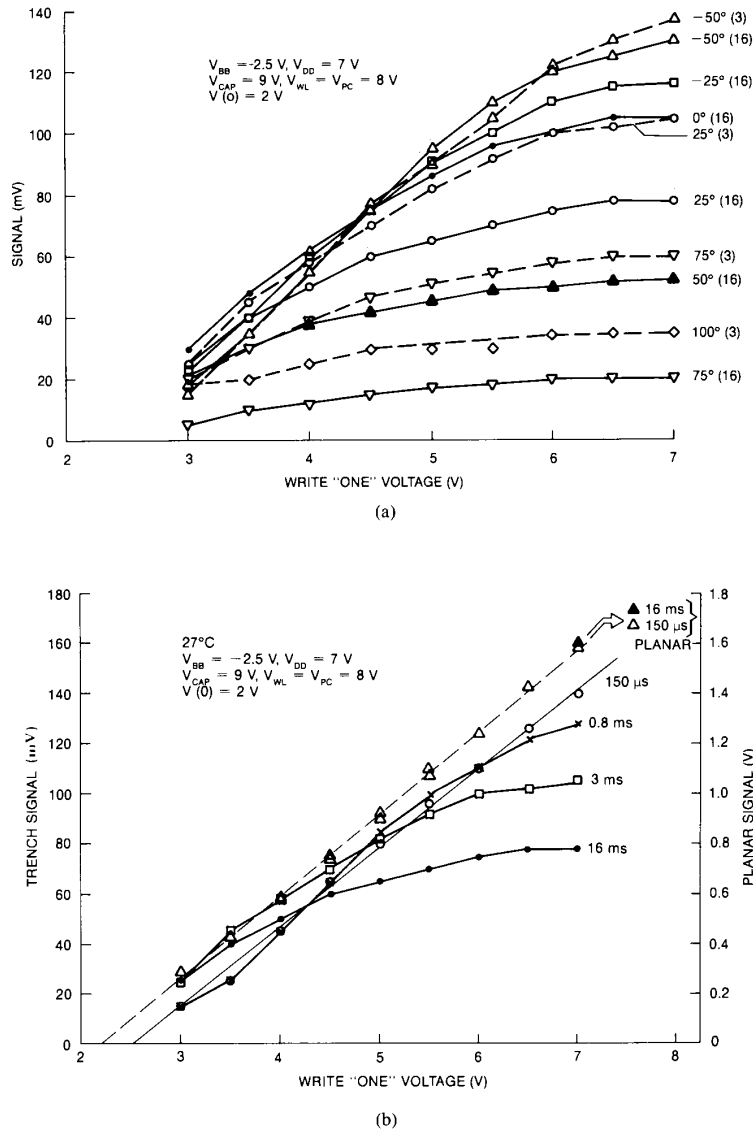


Fig. 4. (a) Output signal dependence on storage voltage level for TTC at various temperatures for 3- and 16-ms refresh. (b) Output level dependence on storage level for TTC and planar cell.

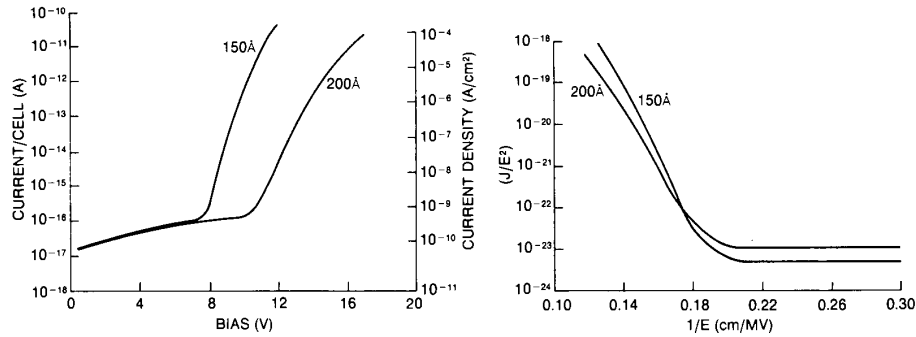


Fig. 5. Fowler-Nordheim plot of leakage through trench capacitor oxide for two different oxide thicknesses.

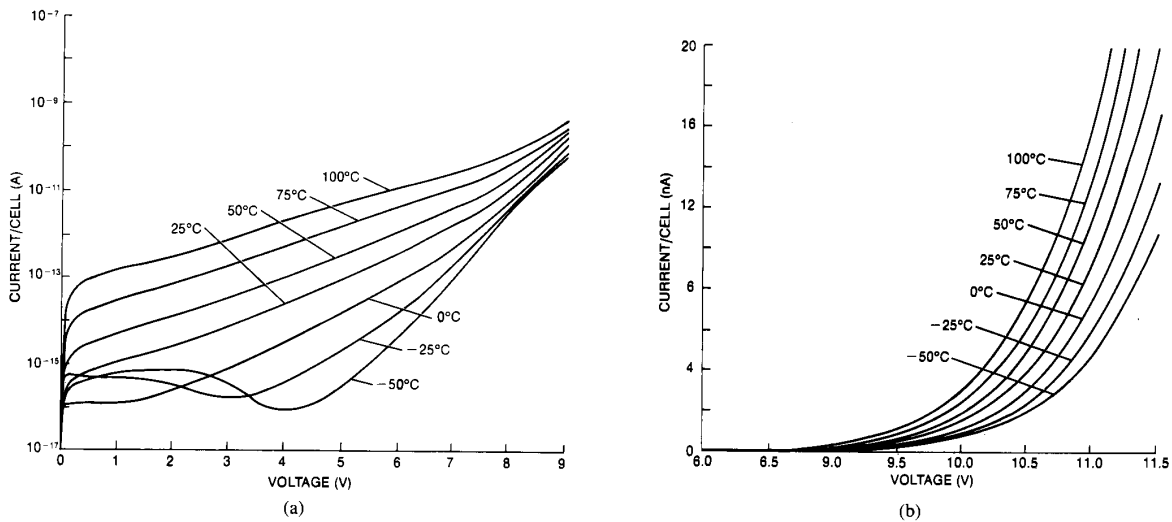


Fig. 6. Temperature dependence of leakage characteristics of TTC at (a) low and intermediate voltages; (b) near-breakdown voltages.

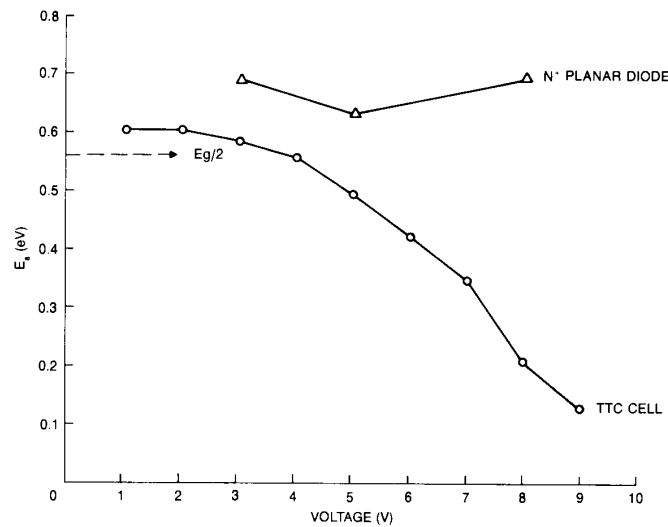


Fig. 7. Activation energy of leakage at different biases.

common buried- n^+ bit line. Typical leakage characteristics are shown for various temperatures in Fig. 6(a). For low voltages, there is a stronger dependence of the leakage on temperature than at higher biases. An Arrhenius' plot of the leakage data yields the activation energy of the leakage current at various voltages stored in the cell (Fig. 7). For comparison, the activation energy is also shown for a planar n^+ diode on the same die. For the n^+ diode, the activation energy E_a is between $1/2 E_g$ and E_g , indicating that the leakage current is a mixture of trap-assisted generation-recombination in the depletion region ($E_a = E_g/2$) and thermal generation of minority carriers ($E_a = E_g$). This attests to the high crystalline quality of the epitaxial layer. However, for the TTC, at low biases, E_a approximately equals $E_g/2$, indicating that generation-re-

combination leakage in the lateral contact junction is the main source in this voltage regime. At higher biases, E_a rapidly reduces indicating some sort of a breakdown phenomena. This could be either due to some tunneling mechanism or due to avalanching at the lateral contact. However, the breakdown characteristics of the TTC (Fig. 6(b)) indicate that the breakdown voltage has a negative temperature coefficient, which is characteristic of tunneling breakdown. This is because at a higher temperature, there is a higher flux of valence electrons, leading to a lower tunneling breakdown voltage. If impact ionization in the high-field region of the lateral contact were responsible, the avalanche breakdown voltage would have a positive temperature coefficient because of the increased optical phonon scattering at higher temperatures. Therefore,

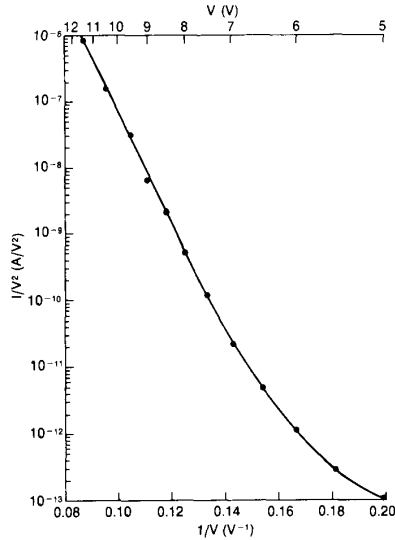


Fig. 8. Fowler-Nordheim plot of leakage current in TTC.

avalanche breakdown of the lateral contact junction can be ruled out; it has to be a tunneling breakdown somewhere.

If the leakage in the TTC at high voltages is indeed due to a Zener mechanism, the leakage current should have the same type of voltage dependence as (2), which is the characteristic form for any tunneling or field emission process. In fact, if the TTC leakage data is replotted as (I/V^2) versus $(1/V)$ (Fig. 8), a linear relation is obtained that indicates a tunneling process. An analysis was done on the leakage data for the actual TTC (Fig. 8) using (2). At each applied voltage, the field in the oxide was calculated and used to estimate the field in the p^+ Si substrate according to (1). A much lower value of $B = 4.3E7$ V/cm was obtained. Using $m^* = 0.198m_0$ for tunneling electron mass in Si [8], a barrier height of 1.2 eV is obtained that reflects the 1.1-eV bandgap of Si.

VI. SIMULATION OF THE TTC

The TTC was simulated using PISCES [9] to study the potential and field patterns around the lateral contact and the storage capacitor. Typical simulation results for two adjacent cells along the wordline are shown in Fig. 9. From the potential contours, it appears that there is no cell-to-cell interaction, and therefore the possibility of cell-to-cell leakage can be discounted.

The potential and electric field profile along the Si-SiO₂ interface is plotted as a function of trench depth in Fig. 10(a), along with the boron concentrations at the corresponding depths. From Fig. 10(a), it is seen that the potential variation along the interface near the lateral contact is not rapid enough to allow tunneling across the p-n junction, i.e., Zener breakdown of the lateral contact itself can be discounted. That is not surprising because if one looks at the boron doping of the substrate near the lateral

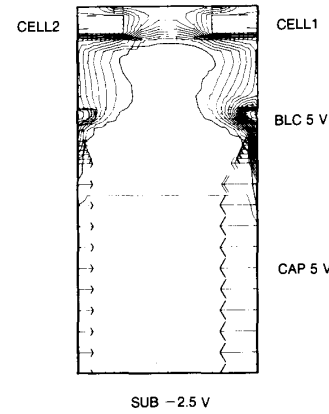


Fig. 9. PISCES simulation of potential contours and electric field distribution in region between two adjoining trench cells along wordline. The cell of interest (CELL1) is at 5 V and the neighboring cell is at 0 V. Electric field magnitudes are proportional to the lengths of the arrows.

contact, it is only about $1E16 \text{ cm}^{-3}$ and that is not high enough to allow Zener breakdown of the junction.

From the potential contours in Fig. 9 it can be seen that there are high electric fields along with sufficient band bending about $1 \mu\text{m}$ below the lateral contact. At a depth of $3.2 \mu\text{m}$ in the trench, where the boron doping is $1E18 \text{ cm}^{-3}$, the electric field is 1.8 MV/cm, and the surface potential changes rapidly with distance from the Si-SiO₂ interface. The midgap potential profile and the electron and hole quasi-Fermi levels are plotted normal to the Si-SiO₂ interface, at a depth of $3.2 \mu\text{m}$, in Fig. 10(b). It is seen that the Si is deep depleted and there is a band bending of $E_g = 1.1 \text{ eV}$ in about $110 \text{ \AA}$, which is enough to create electron-hole pairs by band-to-band tunneling. For higher doping ($1E19 \text{ cm}^{-3}$), the surface electric fields are higher. However, the substrate now acts more and more like a metallic plate and most of the voltage is dropped across the capacitor oxide and very little across the Si. Therefore, the band bending is less than a bandgap in the Si and there can be no tunneling. For lower doping ($1E17 \text{ cm}^{-3}$), although there is adequate band bending, the electric fields are too low because the depletion regions are too wide, i.e., the band bending occurs over too large a distance to allow tunneling breakdown. Therefore there is an intermediate doping region between the high and the low doping where both the criteria for band-to-band tunneling are satisfied. There is a band bending of more than a bandgap and this occurs in a narrow depletion region. If the lateral contact were not tied to the poly gate, the p^+ substrate would go into inversion before allowing a bandgap of band bending, as in a conventional inversion-type DRAM cell.

VII. ESTIMATE OF TUNNELING CURRENT

For the MOS system in deep depletion, the gate voltage V_g drops partly across the oxide V_{ox} and partly across the semiconductor ψ_s .

$$V_g = V_{fb} + V_{ox} + \psi_s \quad (3a)$$

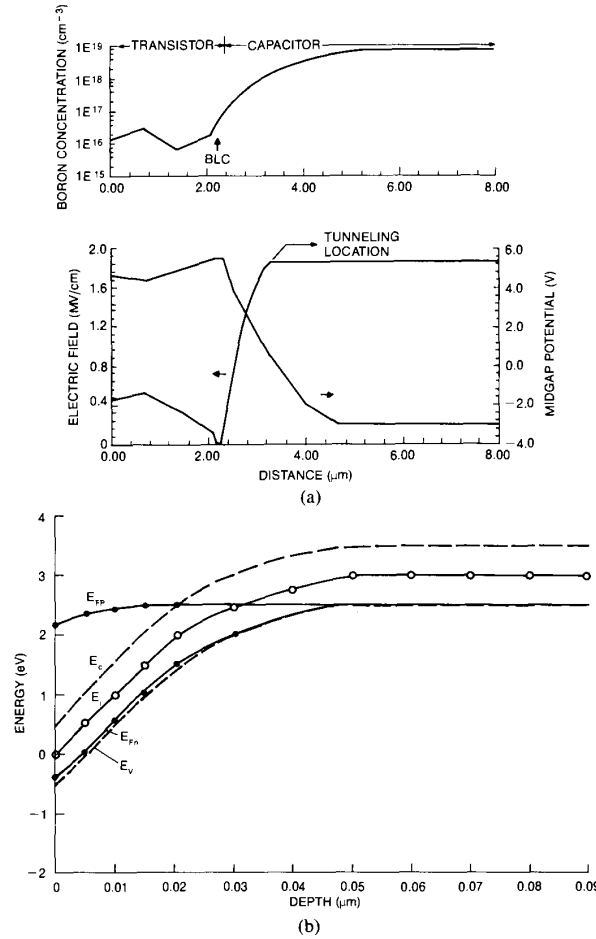


Fig. 10. (a) Potential and electrical field profile along silicon-silicon dioxide interface for CELL I in Fig. 5. (b) Potential and imref variation with distance normal to the interface at depth of 1 μm below lateral contact.

where $V_{fb} = (\phi_{ms} - Q_{ox}/C_{ox})$ is the flat-band voltage.

$$V_{ox} = Q_d/C_{ox} = qN_a W t_{ox}/\epsilon_{ox}$$

and

$$\psi_s = qN_a W^2/(2\epsilon_{si})$$

where W is the depletion width. Therefore

$$V'_g = V_g - V_{fb} = \frac{1}{C_{ox}} \sqrt{(2\epsilon_{si} q N_a \psi_s)} + \psi_s \quad (3b)$$

which gives

$$\psi_s = \left[\sqrt{\left(\frac{\epsilon_{si} q N_a}{2C_{ox}^2} + V'_g \right)} - \sqrt{\left(\frac{\epsilon_{si} q N_a}{2C_{ox}^2} \right)} \right]^2 \quad (3c)$$

A fixed charge density of $1\text{E}11 \text{ cm}^{-2}$ [4] yields a V_{fb} of about -1 V . For a $150\text{-}\text{\AA}$ capacitor oxide, with 5 V stored

in the cell at -2.5-V substrate bias

$$\psi_s = \left[\sqrt{8.5 + (1.58E - 18)N_a} - (1.26E - 9)\sqrt{N_a} \right]^2 \quad (3d)$$

From the above equation, it is seen that ψ_s monotonically decreases with increasing doping N_a . For instance, at $N_a = 1\text{E}17 \text{ cm}^{-3}$, $1\text{E}18 \text{ cm}^{-3}$, and $1\text{E}19 \text{ cm}^{-3}$, ψ_s is 6.5, 3.7, and 0.91 eV, respectively. Thus, for 5 V in the cell, there is insufficient band bending at $1\text{E}19 \text{ cm}^{-3}$, although enough band bending (and hence tunneling) would occur even at $1\text{E}19 \text{ cm}^{-3}$ for higher stored voltages.

To estimate the tunneling current, the WKB approximation is used to compute the tunneling probability P through the barrier [10].

$$P = \exp \left[-2 \int_{x_1}^{x_2} k(x) dx \right] \quad (4a)$$

where $k(x)$ is the wave vector of the tunneling electron in the barrier, and x_1 and x_2 are the classical turning points. The tunneling barrier in Si may be approximated by a triangular barrier of height $E_g = 1.1 \text{ eV}$ [10]. Then

$$P = \exp \left[-\frac{4\sqrt{2m^*}E_g^{1.5}}{3q\hbar E} \right] \quad (4b)$$

Here, the tunneling effective mass in the barrier is for tunneling in the (100) direction from the light hole band to the (110) or X conduction band minima [8].

$$m^* = 2 \left[\frac{1}{3m_l^*} + \frac{2}{3m_t^*} + \frac{1}{m_{lh}^*} \right]^{-1} = 0.198m_0. \quad (5)$$

Following the formulation of Esaki [11], the tunnel current is given by

$$I = A \int [F_c(E) - F_v(E)] P N_c(E) N_v(E) dE \quad (6a)$$

where $N_c(E)$ and $N_v(E)$ are the density of states in the conduction band and valence band and $F(E)$ is the Fermi-Dirac distribution function involving the carrier imrefs (because the TTC is in nonequilibrium). From Fig. 10(b), the average tunneling electric field is 1 MV/cm . From the electron and hole quasi-Fermi levels, it is obvious that practically all the valence band states are full and all the conduction band states are empty. Therefore, the above expression for the tunnel may be considerably simplified as [12]

$$I = AqNvP \quad (6b)$$

which is the tunneling current through a triangular barrier upon N valence band electrons impinging upon the barrier at velocity v . To first order, N may be taken to be the valence band electron density of $2\text{E}23 \text{ cm}^{-3}$ and v to be the thermal velocity of $1\text{E}7 \text{ cm/s}$, giving an electron flux of $Nv = 2\text{E}30 \text{ cm}^{-2} \cdot \text{s}^{-1}$. From Fig. 10(a), it is seen that there is a $1\text{-}\mu\text{m}$ -wide region about $1 \mu\text{m}$ below the lateral contact where tunneling can occur. This corresponds to doping densities between $1\text{E}18$ and $5\text{E}18 \text{ cm}^{-3}$.

Since the trench perimeter at this depth is about $4\text{ }\mu\text{m}$, the tunneling area $A = 4\text{E-}8\text{ cm}^2$. Using $E = 1\text{ MV/cm}$, $P = 6\text{E-}16$. Therefore, the tunnel current per TTC at 5 V (with -2.5-V substrate bias) is

$$I = (4\text{E-}8)(1.6\text{E-}19)(2\text{E}30)(6\text{E-}16) = 7.7\text{ pA/cell.}$$

For comparison, the experimentally observed leakage at 5 V is about 4 pA/cell .

VIII. CONCLUSION

Various leakage mechanisms have been identified in the TTC. Of these, mechanisms such as subthreshold leakage through the trench transistor, cell-to-cell leakage, or Fowler-Nordheim leakage through the trench capacitor oxide itself are insignificant. At low voltages, the main leakage is trap-assisted generation-recombination leakage at the reverse-biased lateral contact. However, at higher stored voltages, band-to-band tunneling in the p^+ Si substrate is the dominant mechanism.

This effect should be operative in the class of trench cells where the charge is stored in a poly plate inside the trench and the outer substrate acts as a capacitor plate. Examples of this type of cell are NEC's buried storage electrode (BSE) cell [13] or IBM's substrate plate cell (SPT) [14]. Of course, in the SPT cell, because of the cell being in an n-tank, the tunneling current should be collected at the n-tank contact instead of the cell itself.

This Zener effect is a very beneficial phenomena for the TTC in terms of long-term reliability. The onset of the tunneling breakdown can be tailored by adjusting the substrate bias or controlling the up-diffused boron profile. Hence, cell functionality is not compromised. By judiciously using this voltage "clamping" mechanism, it should be possible to avoid electrically overstressing the capacitor oxide thereby improving the long-term reliability of the cell.

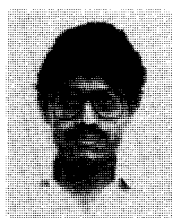
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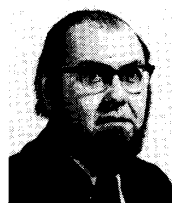
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